Fun with Arduino and Semaphore Signalling – A Radio Scouting Project

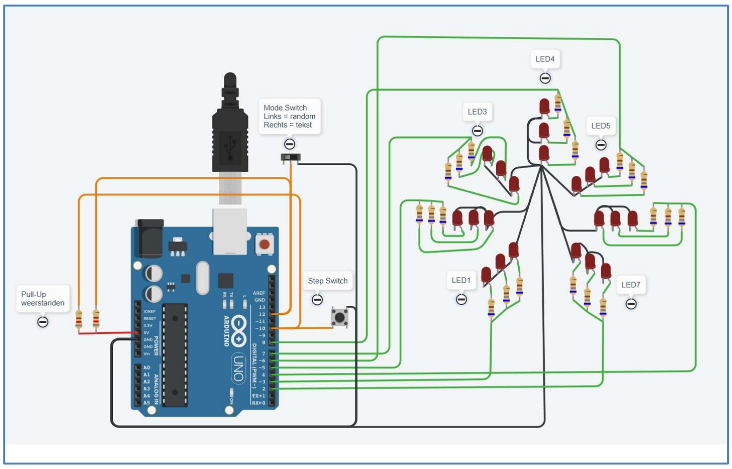
For many radio amateurs, one of the most rewarding parts of the hobby is sharing technical curiosity with young people. This Arduino semaphore project was created with exactly that spirit in mind. It combines electronics, programming, radio history, and practical scouting use into a single small device that is both educational and genuinely fun to build and use. The end result is a compact semaphore trainer box that can be used during scouting meetings and events such as JOTA and TDOTA. It offers a playful way to introduce communication principles while also giving young builders the satisfaction of creating something real and working with their own hands.

The Radio Scouting Fellowship of PA3EFR/J developed this small learning box, which could be used at meetings or as educational decoration.

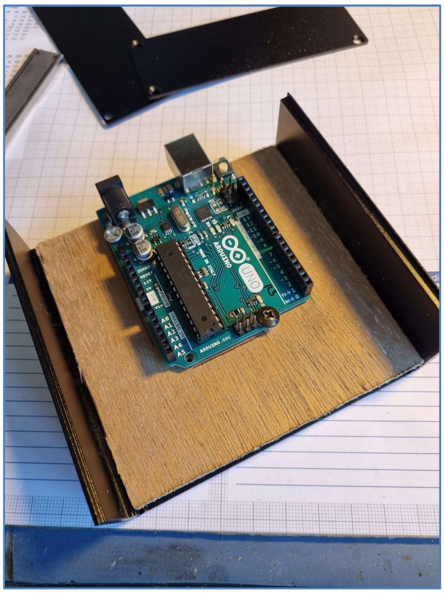
A Project That Is Simply Fun

What makes this project special is how quickly it becomes engaging. During construction, participants enjoy soldering the LEDs, wiring the controls, and watching the circuit slowly come to life. When the software is uploaded to the Arduino, the project immediately starts responding to button presses, which gives instant feedback and keeps motivation high. At the same time, the project introduces the historic semaphore signalling system in a playful way. Instead of learning it from a book, young people see letters appear in front of them and can try to recognize or reproduce them. Once finished, the device turns into a teaching tool that can be used again and again during scouting activities.

The Semaphore Trainer



The semaphore trainer is built around a standard Arduino board and uses seven LEDs to represent the possible arm positions of a semaphore operator. A push button allows the user to step through characters, while a switch selects between operating modes: pre-defined text or random characters. The device can display letters, numbers (with “start-of-numers” and “end-of-numbers” characters built-in); predefined messages (with “strat-of-message-sequence” and “end-of_message_sequence” built-in), and an idle routine where even then characters are included. The goal is not only to show semaphore characters but also to help users actively learn and recognize them.



Learning Without Feeling Like a Lesson

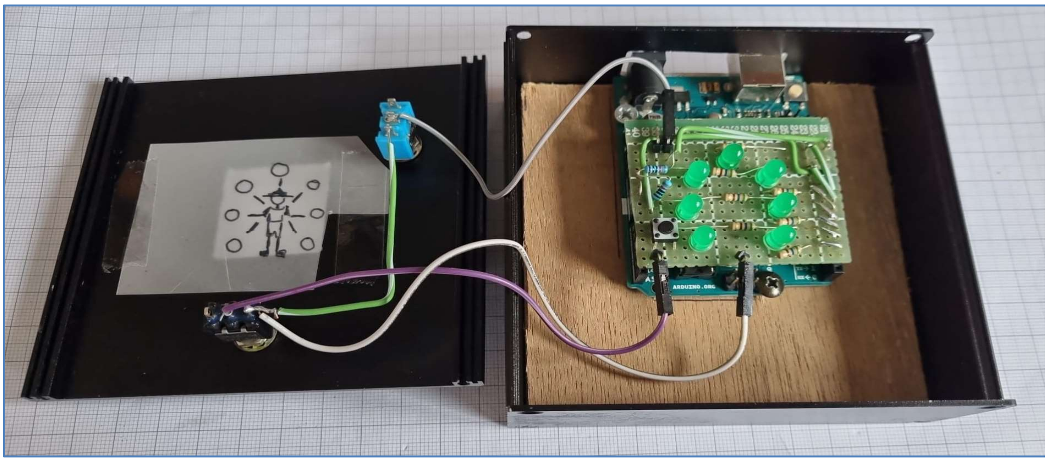
One of the strengths of this project is that learning happens naturally. While building the device, participants are introduced to basic electronics concepts such as polarity, current limiting resistors, and wiring techniques. Programming the Arduino adds another layer by showing how software controls hardware. At the same time, the historical background of visual signalling methods can be explained, linking modern microcontroller technology to traditional communication methods. Because participants are busy building something interesting, the educational aspect feels natural rather than forced.

Simple Hardware, Big Impact

The hardware design intentionally stays simple so that beginners can successfully complete the project. The limited number of components keeps costs low and reduces build errors. Despite this simplicity, the finished device looks impressive and clearly demonstrates how a microcontroller can control multiple outputs in a meaningful way. This balance makes the project suitable for short workshops, club evenings, or multi-session scouting build activities.

Software That Supports Training and Demonstration

The software was written to support both demonstration and training use. The device can show predefined text, generate random training characters, and perform automatic test sequences at startup. Special care was taken to make operation reliable and predictable, which is important when the device is used by many different people during events. The random training mode is designed so that letters appear more often than numbers, making it easier for beginners to learn step by step.



Perfect for JOTA and TDOTA Activities

During radio scouting events, the semaphore trainer provides an excellent additional activity. It can be used to introduce communication history before radio contacts are made, or as an educational game while participants wait for their turn on the radio. The device also helps demonstrate that communication does not always require modern digital technology. Because the trainer is small and robust, it is well suited for outdoor and field use.

Building Together

When building the project with young people, it works well to first introduce the idea of semaphore signalling and explain why it was historically important. After that, construction can be done step by step, allowing participants to see progress quickly. Uploading the software together is usually a highlight moment, because the device immediately starts working. The final stage can focus on learning real semaphore messages and creating small communication games.

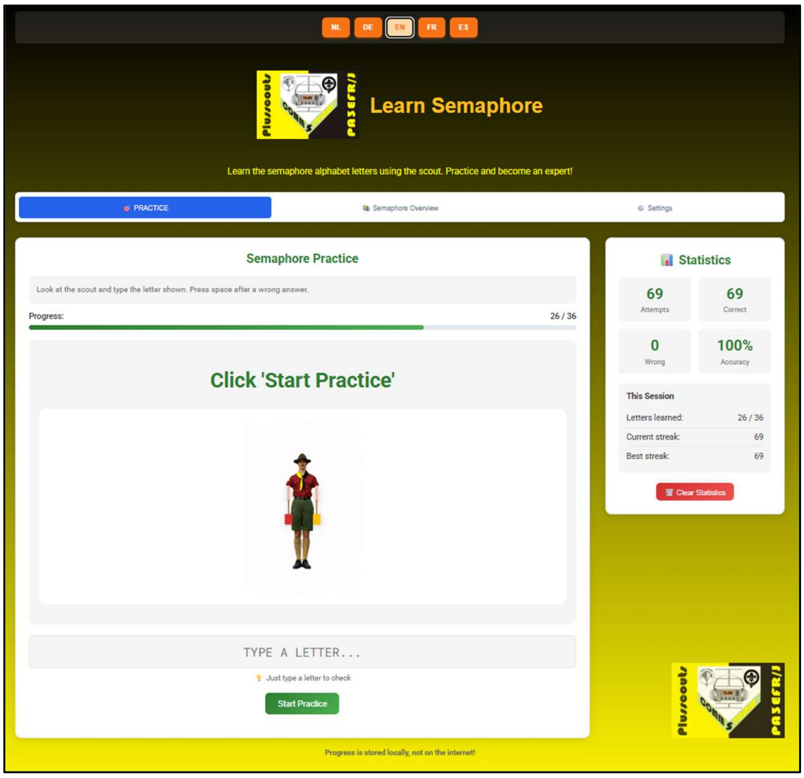
Room for Creativity and Expansion

Although the base project is simple, it leaves plenty of room for creative extensions. More advanced participants can experiment with adding sound feedback, connecting the device to radio equipment, or building larger demonstration models. The Arduino platform makes it easy to modify the software, which encourages experimentation and deeper learning. The repository for this activity: https://github.com/PA3EFR/Arduino_Semafoor where the script can be found for this project.

Conclusion

This Arduino semaphore trainer combines the spirit of amateur radio, education, and fun in a single project. It shows young people that technology can be creative, interactive, and meaningful. The combination of building, programming, and learning a real communication method makes it a powerful learning tool for scouting groups and radio clubs alike.

In case this hardware learning tool is too much or you would like to add an online version of this functionality, then have a look at our site https://qrz.pa3efr.nl/Semafoor/index.html



On behalf of the Plusscouts PA3EFR/J: enjoy building and experimenting.

73, Erwin, PA3EFR